Generalized Split-Explicit Runge–Kutta Methods for the Compressible Euler Equations

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ABSTRACT

The compressible Euler equations exhibit wave phenomena on different scales. A suitable spatial discretization results in partitioned ordinary differential equations where fast and slow modes are present. Generalized split-explicit methods for the time integration of these problems are presented. The methods combine explicit Runge–Kutta methods for the slow modes and with a free choice integrator for the fast modes. Order conditions for these methods are discussed.

Construction principles to develop methods with enlarged stability area are presented. Among the generalized class several new methods are developed and compared to the well-established three-stage low-storage Runge–Kutta method (RK3). The new methods allow a 4 times larger macro step size. They require a smaller integration interval for the fast modes. Further, these methods satisfy the order conditions for order three even for nonlinear equations. Numerical tests on more complex problems than the model equation confirm the enhanced stability properties of these methods.

1. Introduction

Dynamical cores of modern numerical weather codes require the numerical solution of the compressible Euler equations. A variety of spatial discretization techniques is available, among them combinations of finite differences and finite volumes. The typical spatial and temporal discretization parameters suggest the application of explicit time integration methods for the advective terms.

Common integration methods in time are split-explicit methods based on different explicit integration methods for the slow advective modes and the fast sound modes. A popular split-explicit method uses a coupling of a three-stage low-storage Runge–Kutta method (RK3) for the slow modes and the forward backward method for the fast modes (Wicker and Skamarock 2002). Versions of that integration scheme are used in modern weather forecast systems like Weather Research and Forecasting Model (WRF), Nonhydrostatic Icosahedral Atmospheric Model (NICAM), or the Consortium for Small Scale Modeling (COSMO). It is applied in there with additional filtering by divergence damping.

We develop here methods with enlarged stability regions compared to established methods in the case when no divergence damping is applied. These methods are designed with respect to the mathematical concept of temporal convergence order that cannot be applied in the presence of divergence damping because the filter parameter depends on the time step size. Thus, we describe a methodology, how to develop methods that are optimal (or optimized) with respect to

- an order concept,
- a stability concept.

The resulting mathematical problem is quite complex and quite difficult to solve. In fact, even for a method with only three stages, we have to deal with

- up to 12 free parameters;
- nine order conditions, eight of them being nonlinear;
- a stability function that is computed as the maximum of the modulus of eigenvalues of matrices that depend on the varying wavenumber;

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- a stability region we want to maximize, but in which sense?

The solution of such an optimization problem is a complex task that requires several runs of optimization software in combination with serendipity. Thus, we have to fix a few additional parameters. We have decided to run our procedure for a setting where the maximum Mach number is fixed, and we fixed this number to be 1/6. Further criteria a described in the end of section 2 and in section 3. Nevertheless, our procedure can be carried over to other settings (other maximum Mach number choices in particular) in order to construct new methods.

The outline of this paper is as follows: in section 2 we review multirate infinitesimal step methods (which constitute an extension of split-explicit methods) as well as the order conditions and the stability concept for partitioned equations, we extend our methods and the order concept to nonautonomous equations, and we finally discuss which criteria should be met by an optimized method. In section 3 we construct optimized methods by two different approaches. Section 4 follows with numerical tests and a discussion and outlook in section 5 conclude the paper.

2. Review of split-explicit methods

a. Multirate infinitesimal step methods

Multirate infinitesimal step methods (MIS) have been introduced in Wensch et al. (2009). They constitute a generalization of the Runge–Kutta-based split-explicit methods (Wicker and Skamarock 1998, 2002). These methods are designed to solve partitioned equations of the following type:

$$y'(t) = f[y(t)] + g[y(t)],$$

where the right-hand side is composed of two wave-type processes: the slow process f and the fast process g.

In the multirate infinitesimal step splitting method at each stage a variable $Z_{ni}(\tau)$ is defined as the exact solution of an equation $Z'(\tau) = f_c + g[Z(\tau)]$ with a frozen f-value f_c from which a stage value is obtained for a fixed time τ_i. For a given time step size Δt the internal stages Y_{ni} are defined by

$$Z_{ni}(0) = y_n + \sum_j \alpha_{ij}(Y_{nj} - y_n), \tag{1a}$$

$$Z'_{ni}(\tau) = \frac{1}{d_i}\left[\frac{1}{\Delta t}\sum_j \gamma_{ij}(Y_{nj} - y_n) + \sum_j \beta_{ij} f(Y_{nj})\right] + g[Z_{ni}(\tau)], \tag{1b}$$

$$Y_{ni} = Z_{ni}(d_i \Delta t). \tag{1c}$$

We formally add a stage number $s + 1$, which defines the final update $y_{n+1} = Y_{n,s+1}$. Thus, the update and the internal stages are treated in a unique way. The coefficient matrices $(\boldsymbol{\alpha})_{ij} = \alpha_{ij}$, $(\boldsymbol{\beta})_{ij} = \beta_{ij}$, $(\boldsymbol{\gamma})_{ij} = \gamma_{ij}$ are strictly lower triangular. We have s computational stages where the differential equation for $Z_{n,i}$ has to be solved because the first stage simply sets $Y_{n,1} = y_n$.

The length of the integration interval in each stage is given by d_i. In an implementation the exact integration procedure for $z' = f_c + g(z)$ is substituted by a integration procedure of sufficiently high order and sufficiently good stability properties using a smaller step size. The applicable small step size is dictated by the stability properties of the method, thus the number of small steps and therefore the computational cost is determined by the total lengths $D = \sum_i d_i$ of the integration intervals. In this case we use the term macro step size for Δt and micro step size for the smaller step size.

In this paper we restrict ourselves to balanced methods where

$$d_i = \sum_j \beta_{ij}. \tag{2}$$

b. Order conditions

The following abbreviations will be used below:

$$\mathbf{c} = (c_1, \ldots, c_{s+1})^{\mathrm{T}}, \quad \mathbf{D} = \mathrm{diag}(d_1, \ldots, d_{s+1}),$$
$$\tilde{\mathbf{c}} = \boldsymbol{\alpha}\mathbf{c}, \quad D = \sum_i d_i,$$
$$\mathbf{1} = (1, \ldots, 1)^{\mathrm{T}}, \quad \mathbf{R} = (\mathbf{I} - \boldsymbol{\alpha} - \boldsymbol{\gamma})^{-1}, \quad \tilde{\mathbf{b}}^{\mathrm{T}} = \mathbf{e}_{s+1}^{\mathrm{T}}\mathbf{R}\mathbf{D}.$$

When $g = 0$ then the MIS method is equivalent to a classic Runge–Kutta method with coefficient matrix $\mathbf{A} = \mathbf{R}\boldsymbol{\beta}$ and nodes $\mathbf{c} = \mathbf{R}\boldsymbol{\beta}11 = \mathbf{R}\mathbf{D}1$. Thus, the order of the underlying Runge–Kutta method gives the order of the MIS method for a purely slow equation.

For the MIS-RK methods order conditions for the system $y' = f(y) + g(y)$ have been derived in Wensch et al. (2009). For order three the classic order conditions for the underlying RK method are obtained:

$$\mathbf{b}^{\mathrm{T}}\mathbf{1} = 1, \quad \mathbf{b}^{\mathrm{T}}\mathbf{c} = 1/2, \quad \mathbf{b}^{\mathrm{T}}\mathbf{c}^2 = 1/3, \quad \mathbf{b}^{\mathrm{T}}\mathbf{A}\mathbf{c} = 1/6, \tag{3}$$

where $\mathbf{b}^{\mathrm{T}}$ denotes as usual the last row of the matrix $\mathbf{A}$. A method satisfying these conditions is called to be of classic order three, if it satisfies the first two conditions, it is of classic order two.

In addition, there is one further condition ensuring order two:

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TABLE 1. Coefficients of the method MIS2 of full order two determined by line-search optimization with the parent method as the starting point.

β_{21}	0.126 848 494 553		
β_{31}, β_{32}	−0.784 838 278 826	1.374 426 752 68	
$\beta_{41}, \beta_{42}, \beta_{43}$	−0.045 672 708 174 9	−0.008 750 822 711 90	0.524 775 788 629
$\alpha_{32}, \alpha_{42}, \alpha_{43}$	0.536 946 566 710	0.480 892 968 551	0.500 561 163 566
$\gamma_{32}, \gamma_{42}, \gamma_{43}$	0.652 465 126 004	−0.073 276 984 945 7	0.144 902 430 420

$$\tilde{\mathbf{b}}^{\mathrm{T}}(\mathbf{c}+\tilde{\mathbf{c}}) = 1. \tag{4}$$

If a method has classic order of at least two and satisfies this condition, it is of full order two. For order three there are four additional order conditions:

$$\tilde{\mathbf{b}}^{\mathrm{T}}(\mathbf{I}+\boldsymbol{\alpha})\mathbf{A}\mathbf{c} = 1/3, \tag{5}$$

$$3\tilde{\mathbf{b}}^{\mathrm{T}}(\boldsymbol{\alpha}+\boldsymbol{\gamma}/2)\mathbf{R}\mathbf{D}(\mathbf{c}+\tilde{\mathbf{c}}) + \tilde{\mathbf{b}}^{\mathrm{T}}\mathbf{D}(\mathbf{c}+2\tilde{\mathbf{c}}) = 1, \tag{6}$$

$$\mathbf{b}^{\mathrm{T}}\mathbf{R}\mathbf{D}(\mathbf{c}+\tilde{\mathbf{c}}) = 1/3, \tag{7}$$

$$\tilde{\mathbf{b}}^{\mathrm{T}}(\mathbf{c}^2+\tilde{\mathbf{c}}^2+\mathbf{c}\cdot\tilde{\mathbf{c}}) = 1. \tag{8}$$

A method having classic order three and satisfying all additional order conditions for orders two and three is called to be of full order three.

The construction of methods that satisfy the full set of nine third-order conditions is undertaken in (Wensch et al. 2009). These methods improve the stability regions of RK3 by a factor of roughly 2. We further show, that for $\boldsymbol{\alpha} = \boldsymbol{\gamma} = 0$ there is no additional order condition for order two. Therefore any classic Runge–Kutta method of order two or higher gives an split-explicit method of at least order two provided the fast term integration procedure is of order two.

c. Nonautonomous systems

We know how to apply our methods to autonomous systems, where the right-hand side does not depend explicitly on time. Our methods are designed to work for these systems, the order theory has been developed for these systems. When time-dependent forcing terms enter the system, the right-hand side will depend explicitly on time [cf. $y' = f(y)$ with $y' = f(t, y)$]. For our methods, it is not even clear, at which intermediate time points the forcing terms have to be evaluated. In section 4 below such an application is used as a test case.

There is a simple way to solve this problem: we transform the nonautonomous system to an autonomous system and apply the method to the autonomous system. Thus, we obtain a way to apply our methods to nonautonomous systems where the theory for autonomous systems remains valid. A differential equation $t' = 1$ is added in order to achieve this. Because we have to split the right-hand side in fast terms and slow terms we assume $t' = f^t + g^t$ with constants f^t, g^t satisfying $f^t + g^t = 1$. We denote the stage values for time by $t_{ni} = T_{ni}(d_i\Delta t)$ where the intermediate variable is denoted by $T_{ni}(\tau)$. The differential equation for T_{ni} is given by

$$\begin{aligned} T'_{ni} &= \frac{1}{d_i\Delta t}\sum_j \gamma_{ij}(t_{ni}-t_n) + \frac{1}{d_i}\sum_j \beta_{ij} f^t + g^t \\ &= \frac{1}{d_i\Delta t}\sum_j \gamma_{ij}(t_{ni}-t_n) + 1. \end{aligned}$$

We observe that the splitting of t' has no influence caused by the balance condition in Eq. (2). Inserting the initial values $T_{ni}(0) = t_n + \sum_j \alpha_{ij}(t_{nj}-t_n)$ we obtain $t_{n,i} = T'_{n,i}(d_i\Delta t)$ to be

$$\begin{aligned} t_{ni} &= t_n + d_i\Delta t + \sum_j (\alpha_{ij}+\gamma_{ij})(t_{nj}-t_n) \\ \Rightarrow t_{ni} &= t_n + c_i\Delta t. \end{aligned}$$

Thus, the intermediate value T_{ni} increases linearly in τ from $t_n + \tilde{c}_i\Delta t$ up to $t_n + c_i\Delta t$ when τ increases from 0 to $d_i\Delta_t$. Assuming that both f, g are explicitly depending on t we obtain the integration procedure in the nonautonomous case:

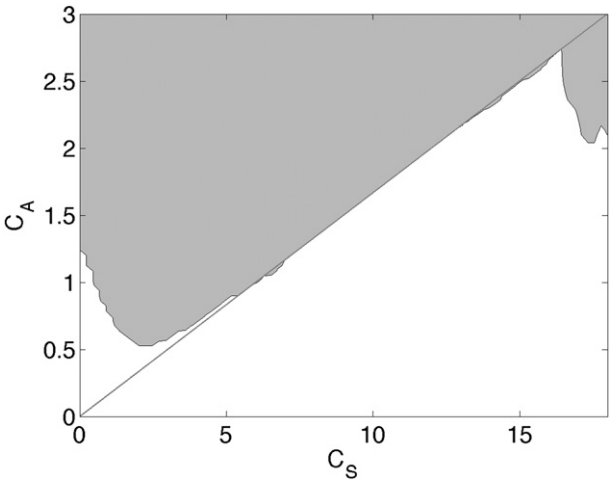

FIG. 1. Stability region of the three-stage method of classic and total second-order MIS2.

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TABLE 2. Coefficients of the method MIS3C of classic order three determined by line-search optimization with the parent method as the starting point.

β_{21}	0.397 525 189 225		
β_{31}, β_{32}	−0.227 036 463 644	0.624 528 794 618	
$\beta_{41}, \beta_{42}, \beta_{43}$	−0.002 952 380 768 40	−0.270 971 764 284	0.671 323 159 437
$\alpha_{32}, \alpha_{42}, \alpha_{43}$	0.589 557 277 145	0.544 036 601 551	0.565 511 042 564
$\gamma_{32}, \gamma_{42}, \gamma_{43}$	0.142 798 786 398	−0.042 891 895 740 2	0.020 272 098 028 2

$$Z'_{ni}(\tau) = \frac{1}{d_i}\left[\frac{1}{\Delta t}\sum_j \gamma_{ij}(Y_{nj} - y_n) + \sum_j \beta_{ij} f(t_n + c_j\Delta t, Y_{nj})\right] + g\left[t_n + \tilde{c}_i\Delta t + \frac{\tau(c_i - \tilde{c}_i)}{d_i}, Z_{ni}(\tau)\right]. \tag{9}$$

d. Split-explicit methods

The splitting methods of Wicker and Skamarock belong to the class above where we have to set $\boldsymbol{\alpha} = \boldsymbol{\gamma} = 0$. The classic RK3 method uses $\beta_{21} = 1/3$, $\beta_{32} = 1/2$, $\beta_{43} = 1$, and vanishing β_{ij} otherwise. The total integration length is $D = \sum_i d_i = 11/6$. The underlying RK method has order three for linear systems, but only order two for nonlinear systems Purser (2007). The splitting method thus has order two when second-order fast term integration is applied.

The methods of Knoth and Wolke (Knoth and Wolke 1998) use nonvanishing α_{ij} where $\alpha_{i+1,i} = 1$, but again the $\boldsymbol{\gamma}$ terms are not present. They analyze the order of the method under the assumption that the exact integration procedure is used. The lowest-order additional order condition they obtain is of order three.

Thus, our methods generalize these methods by allowing for arbitrary α values and additional γ values.

Further, methods based on multistep–multistage methods as slow process integration procedure have been constructed in Jebens et al. (2009). These methods have a much larger stability region than the RK-based methods.

e. Stability

We review shortly the standard procedure to study the stability properties of split-explicit methods (Skamarock and Klemp 1992; Wicker 2009), in particular MIS-RK methods (Wensch et al. 2009). A spatially discretized version of the linearized, continuous, one-dimensional compressible inviscid equations serves as the model equation:

$$\begin{aligned} u_t + Uu_x &= -c_s\pi_x \\ \pi_t + U\pi_x &= -c_s u_x, \end{aligned} \tag{10}$$

where c_s is the speed of sound, U is the constant advection velocity, and π is the Exner pressure. The system exhibits waves propagating with velocities $U \pm c_s$.

The spatial discretization is done on a staggered C grid with grid width Δx by central differencing of the fast acoustic terms and third-order upwind differencing of the advective term (see Skamarock and Klemp 1992; Wensch et al. 2009).

A step of the method applied with fixed Courant–Friedrichs–Lewy (CFL) numbers $C_A = U\Delta t/\Delta x$ and $C_S = c_s\Delta t/\Delta x$ has the following form:

$$\begin{pmatrix} u \\ \pi \end{pmatrix}^{n+1} = S(C_A, C_S, k)\begin{pmatrix} u \\ \pi \end{pmatrix}^{n}.$$

The stability region of a method is then the subset of the (C_S, C_A) plane ($C_S, C_A \geq 0$) where the spectral radius $\rho(S)$ of the amplification matrix S is less or equal to 1 for all wavenumbers. For problems in atmospheric sciences the speed of sound c_s does not vary very much, but the advective velocity varies between zero and a maximum value $U = \mu c_s$. Then, in the (C_S, C_A) plane the critical region is the triangular region defined by

$$0 \leq C_S \leq C_{S,\max}, \quad 0 \leq C_A \leq \mu C_S. \tag{11}$$

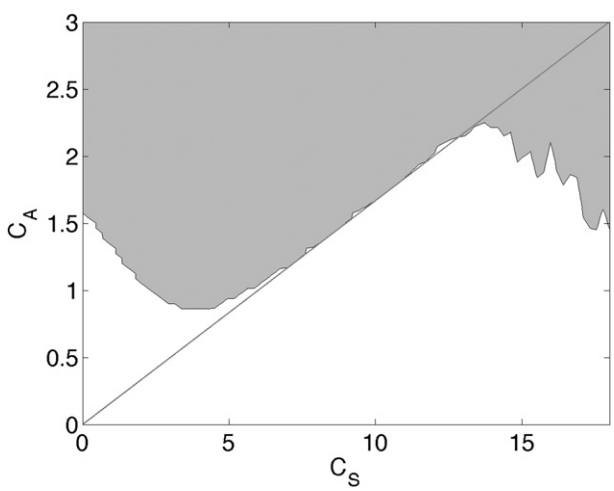

FIG. 2. Stability region of the three-stage method of classic third-order MIS3C.

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TABLE 3. Coefficients of the method MIS4 with full order three and four stages.

β_{21}	0.136 296 478 423		
β_{31}, β_{32}	0.280 462 398 979	−0.016 035 133 359 6	
$\beta_{41}, \beta_{42}, \beta_{43}$	0.904 713 355 208	−1.040 111 831 54	0.652 337 563 489
$\beta_{51}, \beta_{52}, \beta_{53}$	0.067 196 984 554 6	−0.365 621 862 610	−0.154 861 470 835
β_{54}	0.970 362 444 469		
$\alpha_{32}, \alpha_{42}, \alpha_{43}$	0.914 092 810 304	1.142 744 173 97	−0.295 211 246 188
$\alpha_{52}, \alpha_{53}, \alpha_{54}$	0.112 965 282 231	0.337 369 411 296	0.503 747 183 119
$\gamma_{32}, \gamma_{42}, \gamma_{43}$	0.678 951 983 291	−1.389 741 640 70	0.503 864 576 302
$\gamma_{52}, \gamma_{53}, \gamma_{54}$	−0.375 328 608 282	0.320 925 021 109	−0.158 259 688 945

We call this region the stability triangle. Further, for tracer advection stability on the line $C_S = 0$ is critical (i.e., the length $C_{A,\max}$ of the interval on the C_A axis where the method is stable).

f. Design criteria

The properties of a method are determined by

- Computational cost: Number of stages s, total length of fast process integration intervals $D = \sum_i d_i$.
- Accuracy: Approximation order of the underlying Runge–Kutta method and the splitting scheme.
- Stability: For fixed μ the size $C_{S,\max}$ of the stability triangle and the maximum tracer CFL number $C_{A,\max}$. Further, stability has to be checked with respect to both the exact integration procedure and the small step integration procedure.

The full set of nine order conditions is rather restrictive. Further, the fast process integration procedure has to be very simple for reasons of computational efficiency. Thus, it will be of order two at most. A very promising candidate is the Störmer–Verlet method.

We therefore impose the following properties:

- The method has $s = 3$ stages.
- The full set of four third-order conditions for the underlying Runge–Kutta method is satisfied because this method is also used for tracer advection.
- The additional order condition for order two for partitioned systems is satisfied.
- We set the ratio $C_A/C_S = \mu = 1/6$.
- The fast integration procedure is Störmer–Verlet.

3. Construction of methods with improved stability region

We have followed two different approaches to construct three-stage MIS methods of at least full order two with improved stability properties. In particular, we construct one three-stage method (MIS2) with full order two and several three-stage methods with classic order three (MIS3C, TVDMIS3A, and TVDMIS3B). In addition, we construct a four-stage method with full order three (MIS4).

a. Genetic optimization

We use a genetic algorithm to optimize the objective function which combines good stability properties with small integration interval D. The initial population consists of $N = 1000$ randomly generated second-order three-stage methods. Second-order accuracy is established by a parameterization of second-order methods. In each optimization step a new population is generated as follows:

1) The method with the best value of the objective function is chosen as parent of the next generation.
2) The coefficients of the parent are randomly perturbed within a perturbation radius R to generate $N - 1$ children.
3) The next generation is formed from the parent and its children.
4) The perturbation radius R is decreased.
5) The algorithm is repeated until there is no improvement in 10 successive steps.

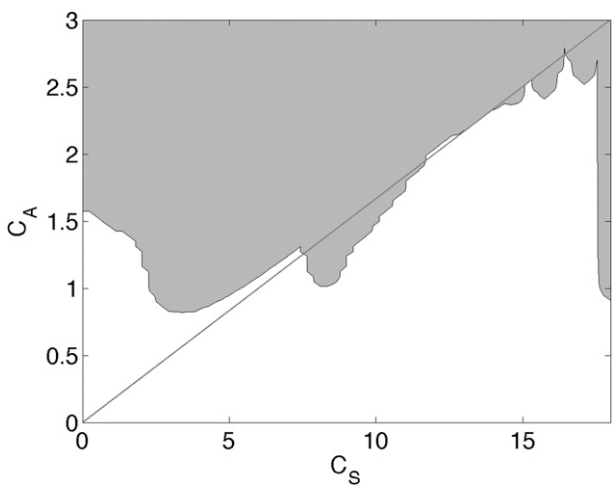

FIG. 3. Stability region of the four-stage method of full third-order MIS4.

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TABLE 4. Coefficients of the method MIS4a with full order three and four stages.

β_{21}	0.387 584 446 414 503 18		
β_{31}, β_{32}	−0.025 318 448 354 142 823	0.386 689 430 873 104 03	
$\beta_{41}, \beta_{42}, \beta_{43}$	0.208 999 835 235 533 25	−0.458 566 484 763 712 31	0.434 231 875 734 257 48
$\beta_{51}, \beta_{52}, \beta_{53}$	−0.461 861 719 563 333 27	−0.461 861 719 563 333 27	0.830 450 621 224 628 09
β_{54}	0.270 149 149 002 503 92		
$\alpha_{32}, \alpha_{42}, \alpha_{43}$	0.523 492 499 223 856 10	1.168 337 436 689 362 9	−0.757 620 802 417 126 37
$\alpha_{52}, \alpha_{53}, \alpha_{54}$	−0.036 477 233 846 797 109	0.569 361 487 307 404 77	0.477 462 630 025 996 81
$\gamma_{32}, \gamma_{42}, \gamma_{43}$	0.131 450 897 962 265 42	−0.368 558 576 487 478 81	0.331 592 326 366 005 50
$\gamma_{52}, \gamma_{53}, \gamma_{54}$	−0.065 767 130 537 473 045	0.040 591 093 109 036 858	0.064 902 111 640 806 712

Note, that the method may converge also for nonsmooth or even discontinuous objective functions.

The objective function combines a large stability area with a minimal length of integration intervals. The primary target is a large stability area. Among all methods where $C_{S,\max}$ is at least 30, the sum of the fast integration intervals $\sum d_i$ is minimized. For the integration of the fast part we use the Störmer–Verlet method with $n_s = 33$ small time steps per large time step. Our objective function φ is given by

$$\hat{\varphi} := \max(0, 30 - C_{S,\max}), \quad \varphi := \begin{cases} \hat{\varphi} & \text{if} \quad \hat{\varphi} > 0 \\ -1/\sum d_i & \text{if} \quad \hat{\varphi} = 0 \end{cases}. \tag{12}$$

The method we have obtained by the genetic optimization process has an extended stability area, but seems to be quite sensitive to the selected number of micro steps.

Therefore, we use this method as a parent method for further line-search-type optimization procedures. In a first step we add constraints guaranteeing a stability triangle of at least $C_{S,\max} \geq 12$ and minimize the total length D of the fast process integration intervals. The coefficients of the resulting method MIS2 are given in Table 1. The stability region is displayed in Fig. 1. The target region is covered almost completely. The size is approximately $C_{S,\max} \approx 17$. The number of small steps is taken to be 18. Now, we add the classic third-order accuracy conditions as additional constraints and continue the line-search optimization procedure. The resulting method MIS3 is given in Table 2. It has classic order three but only full order two. The corresponding stability region is depicted in Fig. 2.

Finally, we tried to satisfy all nine order conditions for full order three for partitioned systems. We did not succeed to find a method with three stages with appropriate stability properties. By allowing for a fourth stage we have constructed such a method (MIS4) satisfying all nine order conditions for order three displayed in Table 3. Its stability region is displayed in Fig. 3. Again, roughly the complete target area is covered. The size is $C_{S,\max} \approx 17$. In addition, we have tried for an optimized method for the case $C_A/C_S = 1/4$. We call this method MIS4a. Its coefficients are given in Table 4, its stability region is displayed in Fig. 4.

b. Methods based on TVD-RK

The underlying RK method is traditionally used in the advection scheme for the transport equations for substances like water, vapor, etc. It is, therefore, natural to consider TVD-RK methods as underlying RK methods. We aim at methods where the overall length $\sum d_i$ of the small time step integration intervals is minimal. For stage 2 we have $d_2 = c_2$. Therefore, methods with large node c_2 will not give low values of D. Thus, the optimal method of Gottlieb and Shu is not appropriate because the first step in stage 2 is of length $d_2 = c_2 = 1$.

We have found out that the method in Table 5 is a promising candidate with $d_2 = c_2 = 2/3$.

In the appendix we show that it is optimal in a certain sense, although the optimal method (Gottlieb and Shu 1998) has a larger total variation diminishing (TVD) radius.

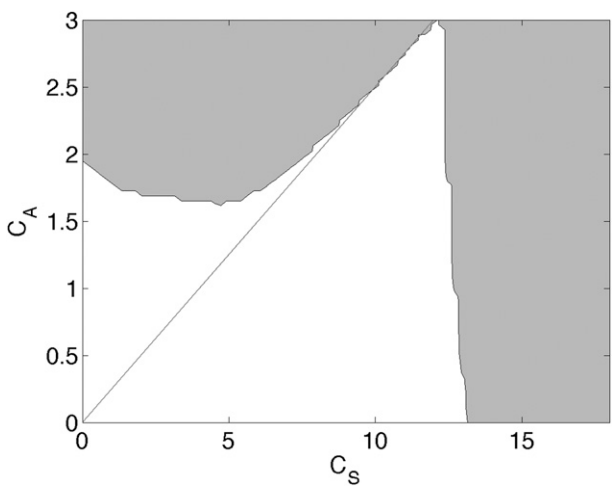

FIG. 4. Stability region of the four-stage method of full third-order MIS4a.

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TABLE 5. The underlying TVD-RK method.

0			
2/3	2/3		
2/3	2/9	4/9	
	1/4	3/16	9/16

Our settings for the optimization procedure fmincon of Matlab are as follows:

For prescribed a_{ij}, our optimization variables are α_{ij} and γ_{ij}—a total of six free parameters. The classic order conditions for order three are always satisfied. The additional order condition for order two is added to our optimization problem as a constraint.

An assumed size $C_{S,\max}$ of the stability area is always as additional parameter present, either as a fixed parameter or an additional optimization variable. For stability we have to assure that the stability function has modulus not greater than one in the triangular stability region defined by Eq. (11). Because of the maximum principle for holomorphic functions it suffices to assure this on the boundary of the triangular region. The modulus of the two eigenvalues of the amplification matrix is evaluated on a set of representative wavenumbers and on a grid on the boundary of the triangular region. These inequalities are always added as constraints to our optimization problem.

The simultaneous minimization of $D = \sum_i d_i$ and $-C_{S,\max}$ is achieved by the successive application of different choices for the objective function and additional free parameters.

1) $C_{S,\max}$ is a free parameter. The objective function is $\phi = -C_{S,\max}$. The D is not taken into account.
2) $C_{S,\max}$ is a fixed parameter. The objective function is $\phi = -D$.
3) $C_{S,\max}$ is a free parameter. The objective function is $\phi = -C_{S,\max}/D$.

We found out that the most promising results have been obtained by applying strategies 1), 2) in turn, whereas in the final optimization steps strategy 3) has been used. The optimizer was matlab's fmincon procedure.

The stability function is evaluated for a finite number of small steps with Störmer–Verlet. The number is determined such that the small step size is within the stability region of the Störmer–Verlet method, whereas an additional safety factor of 0.9 is applied.

The coefficients of two methods, named TVDMISA and TVDMISB are given in Tables 6 and 7.

Both methods have a stability area size of $C_{S,\max} \approx 12$. The total small step integration length is $D \approx 1.25$. If a macro step with $C_S = 12$ is applied, the suggested small step numbers are nss = 8, 2, 5. In our experiments we found out that the method works nicely if we round this up to small step numbers nss = 9, 3, 6. By taking 18 small steps for $C_S = 12$ this corresponds in fact to $D = 1.5$.

Below we show the stability regions for the methods MIS2, MIS3C, MIS4, MIS4a, TVDMISA, and TVDMISB in Figs. 1 to 6.

All six methods have an exceptionally extended stability region. In fact, we could use CFL numbers almost up to 18 with respect to the speed of sound (although this results in an increased number of small steps in the fast term integration procedure, these steps are cheap compared with the advection procedure). An overview of the methods constructed here is given in Table 8 where RK3 is added as a reference.

4. Numerical tests

Our methods have been designed with respect to a stability model based on the linear acoustics equation. At this point, we test our methods on a selection of examples from the literature that serve to evaluate further qualitative and quantitative properties. In particular, we evaluate scale-selctive damping properties (Vater et al. 2011), order of convergence for nonlinear problems (Durran and Blossey 2012), and the performance on the rising bubble benchmark problem (Wicker and Skamarock 1998) as well as on the cold bubble downburst problem (Straka et al. 1993).

a. Scale-selective damping properties of the MIS method

The damping properties are important with respect to the time integration methods for compressible atmospheric flow with respect to highly oscillatory spatial modes. These modes may be excited by local small-scale heat release (see Vater et al. 2011). They can be removed by implicit time integration methods but then the smooth modes will be damped too. In Vater et al. (2011) blended methods are constructed that damp the short wave modes only but minimize the amplitude and dispersion error for long wave modes. In this section we

TABLE 6. Coefficients of method TVDMISA.

d_1, d_2, d_3	0.666 666 666 666 666 6	0.161 972 697 409 560 9	0.431 346 556 198 713 3
$\alpha_{32}, \alpha_{42}, \alpha_{43}$	0.194 636 060 564 745 7	0.397 120 013 678 661 4	0.260 943 460 621 180 1
$\gamma_{32}, \gamma_{42}, \gamma_{43}$	0.562 404 893 320 912 9	0.440 846 747 571 327 7	−0.245 930 056 169 239 1

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TABLE 7. Coefficients of method TVDMISB.

d_1, d_2, d_3	0.666 666 666 666 666 6	0.189 515 853 443 662 3	0.412 215 672 454 775 8
$\alpha_{32}, \alpha_{42}, \alpha_{43}$	0.426 682 328 633 109 8	0.265 707 790 161 737 7	0.414 899 668 918 666 9
$\gamma_{32}, \gamma_{42}, \gamma_{43}$	0.289 043 891 201 396 6	0.451 135 600 713 342 1	−0.250 066 568 475 910 2

apply the MIS method to their test problem and show the scale-selective damping properties of the MIS methods.

The MIS method reduces to a classic Runge–Kutta method if the fast part g is absent. Here, we consider the opposite case where f is absent. If the exact solution operator is applied to the differential equation for Z_{ni} at each stage, we obtain the exact solution of the differential equation $y' = g(y)$ in the case of α_{ij} and $\gamma_{ij} = 0$. This is especially the case for the RK3 method; thus, scale-selective damping properties cannot be obtained.

For methods with α_{ij} and $\gamma_{ij} \neq 0$ there is not necessarily an exact integration of $y' = g(y)$ performed, even if the exact solution operator is applied in each stage. Thus, these methods may show scale-selective damping.

The linearized one dimensional Euler equation in Eq. (10) serves as a test equation with CFL number 10 and background velocity $U = 0$. In the test case from Vater et al. (2011) a high- and a low-frequency signal are overlaid by amplitude modulation. The exact solution is a signal traveling with constant velocity. A numerical discretization should in the ideal case

- resolve long waves accurately,
- filter inaccurately represented wavelength fast.

In Fig. 7 we display the initial solution and the numerical results given by the MIS method when Störmer–Verlet is applied as fast term integrator. The number of small steps is chosen such that the stability requirements for this method are met. In Fig. 8 the results from RK3 are given: the top image shows the results with Störmer–Verlet as fast term integrator. The bottom image shows the results when in addition divergence damping is applied.

For RK3 the damping properties of the fast term integration procedure determine the damping properties of the method. There is no damping of the short-wave modes. This can be achieved by additional divergence damping, which gives satisfactory results too. The MIS methods exhibit scale-selective damping properties without divergence damping.

b. Numerical order tests

Besides the theoretical prediction of the order of our methods in case of exact or finite small step integration a numerical test of the achieved orders of temporal convergence seems appropriate now.

In this section we evaluate the performance of split-explicit methods in a nonlinear test case. We give a shortened description here taken literally from the original contribution Durran and Blossey (2012).

The test case involves two-dimensional (x–z) nonlinear gravity waves generated by a localized region of nondivergent forcing in a stratified shear flow. With spatial coordinates given in kilometers, the background horizontal wind ($\mathrm{m\,s^{-1}}$) is

$$u_0(z) = 5 + z + 0.4(5 - z)(5 + z).$$

The waves are forced by the curl of a nondivergent streamfunction ($\mathrm{m^2\,s^{-1}}$):

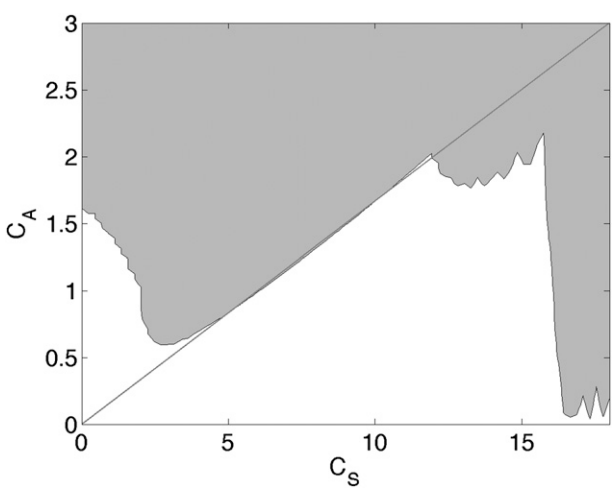

FIG. 5. Stability region of the TVD-based method TVDMISA.

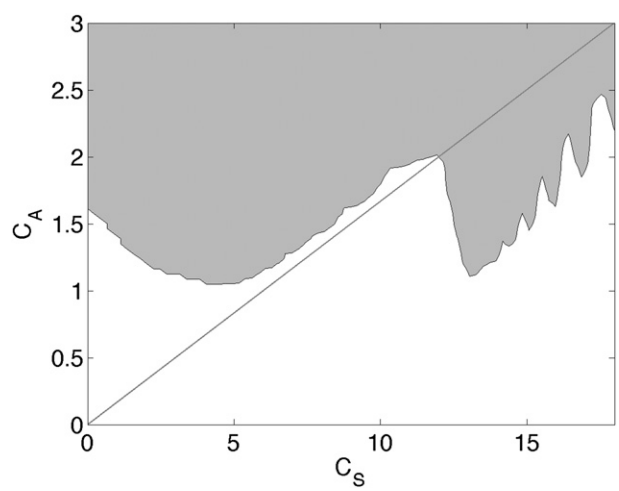

FIG. 6. Stability region of the TVD-based method TVDMISB.

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TABLE 8. Properties of various MIS methods (boldface indicates where MIS4 is stable for $C_{S,\max} = 16$ only for Mach numbers slightly $<1/6$).

Method	Stages	Classic order	Full order	$C_{S,\max}$	$D = \sum_i d_i$
MIS2	3	2	2	16	1.19
MIS3C	3	3	2	13	1.22
MIS4	4	3	3	**8(16)**	1.43
MIS4a	4	3	3	12	1.47
TVDMISA	3	3	2	12	1.26
TVDMISB	3	3	2	12	1.27
RK3	3	2	2	3.14	1.83

$$\psi(x,z,t) = \psi_0\left(\frac{\pi x}{L_x}\right)\sin(\omega t)\exp\left[-\left(\frac{\pi x}{L_x}\right)^2 - \left(\frac{\pi z}{L_z}\right)^2\right].$$

This example is dominated by nonhydrostatic motion and employs a grid spacing characteristic of high-resolution convective cloud models. It is governed by the following system:

$$\begin{aligned}
\frac{D}{Dt}u + \frac{\partial P}{\partial x} &= -\frac{\partial \psi}{\partial z} + \frac{u_0(z) - \overline{u}(z,t)}{\tau},\\
\frac{D}{Dt}w + \frac{\partial P}{\partial z} &= b + \frac{\partial \psi}{\partial x},\\
\frac{D}{Dt}b + N^2 w &= 0,\\
\frac{D}{Dt}P + c_s^2\left(\frac{\partial u}{\partial x} + \frac{\partial w}{\partial z}\right) &= 0.
\end{aligned}$$

We will investigate convergence in the time domain while keeping the spatial discretization fixed. In all simulations the numerical domain is horizontally periodic and bounded by flat rigid upper and lower surfaces. Let m and n be integer indices; the mesh is staggered so that the equation for P applies at points $(m\Delta x,\ n\Delta y)$, the equation for u applies at points $[(m-1/2)\Delta x, n\Delta z]$, and the equations for b and w apply at points $[m\Delta x, (n-1/2)\Delta z]$. Using the operator notation

$$\begin{aligned}
\delta_{nx} f(x) &= \frac{f(x + n\Delta x/2) - f(x - n\Delta x/2)}{n\Delta x},\\
\langle f(x)\rangle^{nx} &= \frac{f(x + n\Delta x/2) + f(x - n\Delta x/2)}{2},
\end{aligned}$$

the spatial discretization has the following form:

$$\begin{aligned}
\frac{\partial u}{\partial t} + \frac{1}{2}\delta_{2x}(u^2) + \langle\langle w\rangle^x \delta_z u\rangle^z - \frac{u_0(z) - \overline{u}(z,t)}{\tau} + K[(\Delta x\delta_x)^2 + (\Delta z\delta_z)^2]^2 u &= -\delta_x P\\
\frac{\partial w}{\partial t} + \langle\langle u\rangle^z \delta_x w\rangle^x + \frac{1}{2}\delta_{2z}(w^2) + K[(\Delta x\delta_x)^2 + (\Delta z\delta_z)^2]^2 w &= -\delta_z P + b\\
\frac{\partial b}{\partial t} + \langle\langle u\rangle^z \delta_x b\rangle^x + \langle\langle w\rangle^z \delta_z b\rangle^z + K[(\Delta x\delta_x)^2 + (\Delta z\delta_z)^2]^2 b &= -N^2 w\\
\frac{\partial P}{\partial t} + \langle u\delta_x P\rangle^x + \langle w\delta_z P\rangle^z &= -c_s^2(\delta_x u + \delta_z w),
\end{aligned}$$

where the discretized operators on the right-hand side form the function g. A simple parameterization of turbulent mixing in a nearly inviscid fluid is imposed through the fourth-derivative hyperdiffusion terms.

To estimate the order of the temporal discretization we follow the methodology applied in Durran and Blossey (2012). A high accuracy solution has been computed with the classic Runge–Kutta method of fourth order with a time step size of 10^{-4} in order to estimate the discretization error. The same spatial discretization is used in all our tests. The time step size (in seconds) varies over the set $\{1, 2.5, 5, 10, 15\}$. The results are displayed in Fig. 9. The ideal slope for a method of order one, two, three is indicated by dotted, slash–dotted, and slashed lines. The method MIS4 with for stages and predicted order three clearly shows third-order error behavior. The methods MIS3C and TVDMISB as well as RK3 show second order of convergence. Finally, we display order results for the method RK3TVD from Baldauf (2010) that has been derived from a third-order TVD method, too. This method also fits in the class of our methods. It displays

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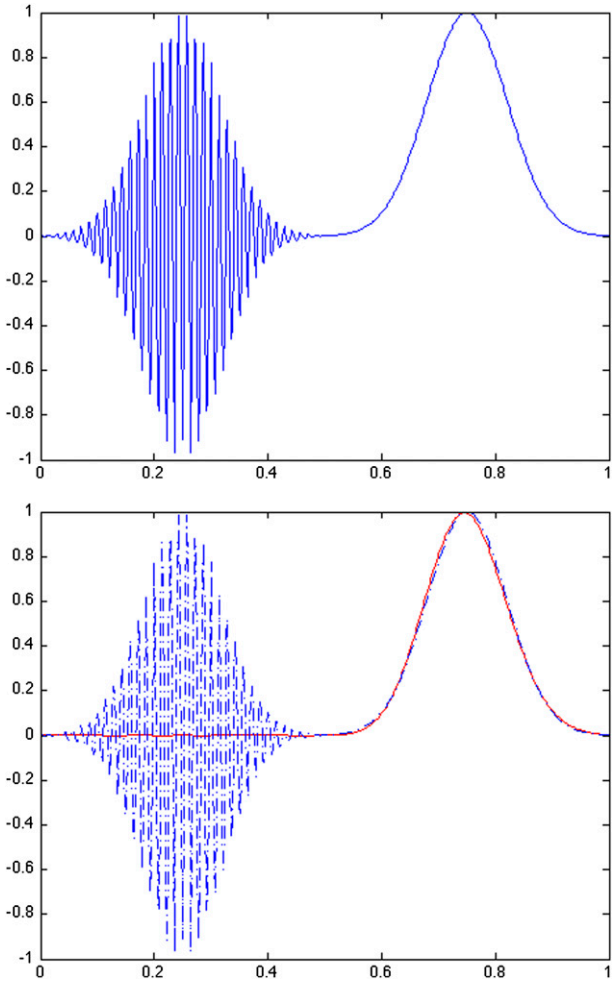

FIG. 7. (top) Exact solution and (bottom) exact solution computed by TVDMISA with Störmer–Verlet.

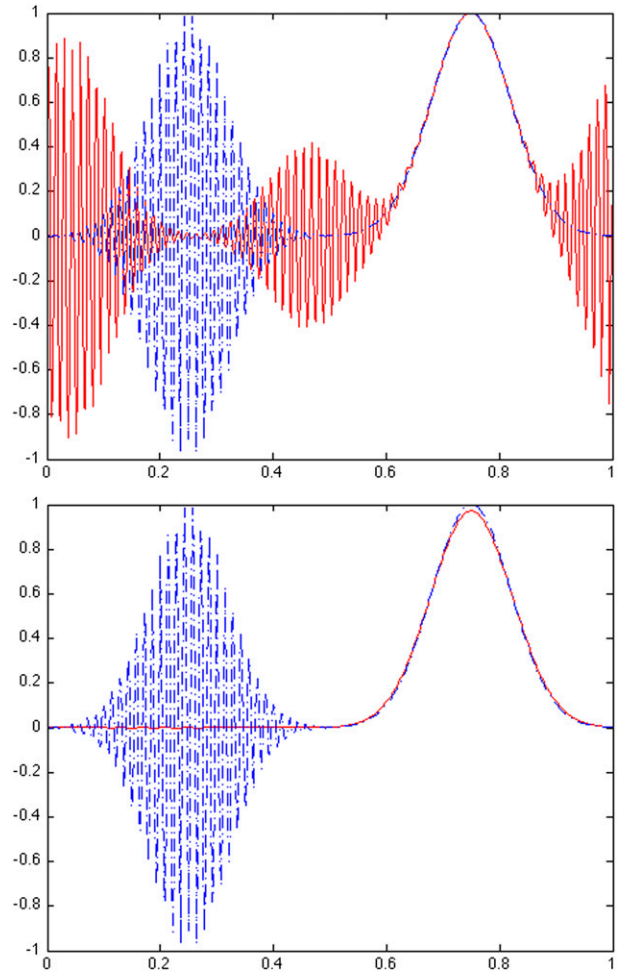

FIG. 8. Classic RK3 (top) with Störmer-Verlet and (bottom) with additional divergence damping.

clearly the first order of convergence. This again reinforces the fact that without a careful order analysis it is difficult to derive higher-order methods of splitting type.

c. *The rising bubble benchmark example*

We apply our methods to the test case from Wicker and Skamarock (1998) where a warm bubble rises driven by buoyancy in an isentropic atmosphere.

The test conditions are exactly the same as in Wensch et al. (2009). We display the results from there (methods RK3 and MIS3A) as well as results on the methods constructed in this paper (MIS3C, MIS4, and TVDMISA) in Table 9. We have solved the problem with the methods above with several different time step sizes. Divergence damping is not applied. The maximum step size for which a stable integration was possible is displayed in Table 9. Whereas MIS3A improves the maximum macro step size by a factor of almost 2, the newly constructed methods allow even 4 to 5 times larger macro step sizes.

d. *The cold bubble downburst benchmark example*

In contrast to the example above we consider now a cold bubble falling down. We use the test example from Straka et al. (1993) with the modifications that have been made in Wicker and Skamarock (2002). On the ground the bubble generates winds of magnitude $30\,\mathrm{m\,s^{-1}}$, several Kelvin–Helmholtz waves are generated. Snapshots of the solution at $t = 0, 450$, and 900 s are displayed in Fig. 10. A grid width of 100 m is used in both directions. With a background wind of $20\,\mathrm{m\,s^{-1}}$ present the maximum Mach number is slightly above 1/6. The MIS integrations schemes have been implemented in the All Scale Atmospheric Model (ASAM). The compressible Euler equations are discretized on a staggered grid, where the advection operator is approximated by third-order upwinding; all other operators are approximated by second-order central differencing. We have solved the problem with a macro step size of $\Delta t = 1, 2,$

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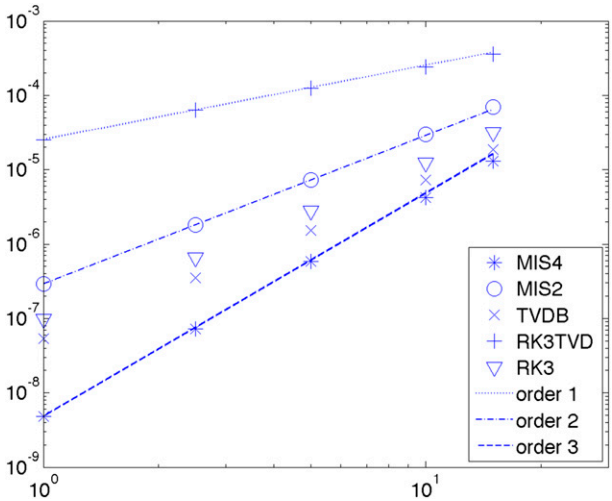

FIG. 9. Error vs step size for a variety of methods.

and 3 s. The corresponding maximum micro step size is 1/6 s. Note, that RK3 failed on these step sizes without divergence damping. Therefore, RK3 is applied in combination with divergence damping [see Wicker and Skamarock (1998), with $\alpha = 0.025$].

Contour plots for the result at $t = 900$ s using a macro step size of $\Delta t = 1$ s are given in Fig. 11. Further, the maximum and minimum values for perturbation potential temperature are displayed in the top left of each diagram . There is a good agreement in these values for all five methods. To evaluate the accuracy we display the deviation from symmetry on the upper right in each diagram. There is also a good agreement between all five methods. The five methods for which the results are displayed are (from top to bottom) RK3, MIS4a, TVDMISB, MIS3C, and MIS2. The properties of TVDMISB, MIS3C, and MIS2 can be reviewed in Table 8. We point out, that the four-stage method MIS4 failed on this example. Instead, we display in the second image the results obtained with another method with four stages designed to be stable for Mach numbers up to 1/4. The other three MIS methods as well as RK3 (with divergence damping) work stable with this step size. We have increased the step size to $\Delta t = 2$ and 3 s. In Fig. 12 we display the results for the latter case. Even with $\Delta t = 3$ s all of the methods work stable, where again divergence damping has to be applied for RK3. Deviations from symmetry are larger than in the case $\Delta t = 1$ s.

TABLE 9. Maximum time step size for the rising bubble test case. All methods are applied without divergence damping.

Method	RK3	MIS3A	MIS3C	MIS4	TVDMISA
Large time step (s)	0.9	1.6	5.0	4.0	4.1

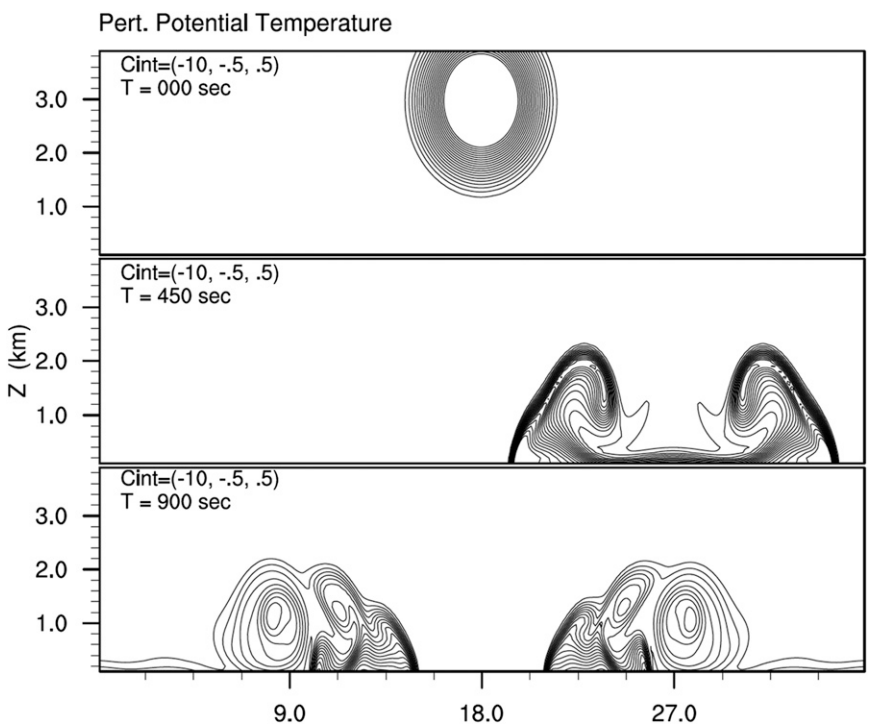

FIG. 10. Potential temperature deviation of background state at the beginning, middle, and end of the simulation. Contours range from −10 to 5 (°C) in increments of 0.5.

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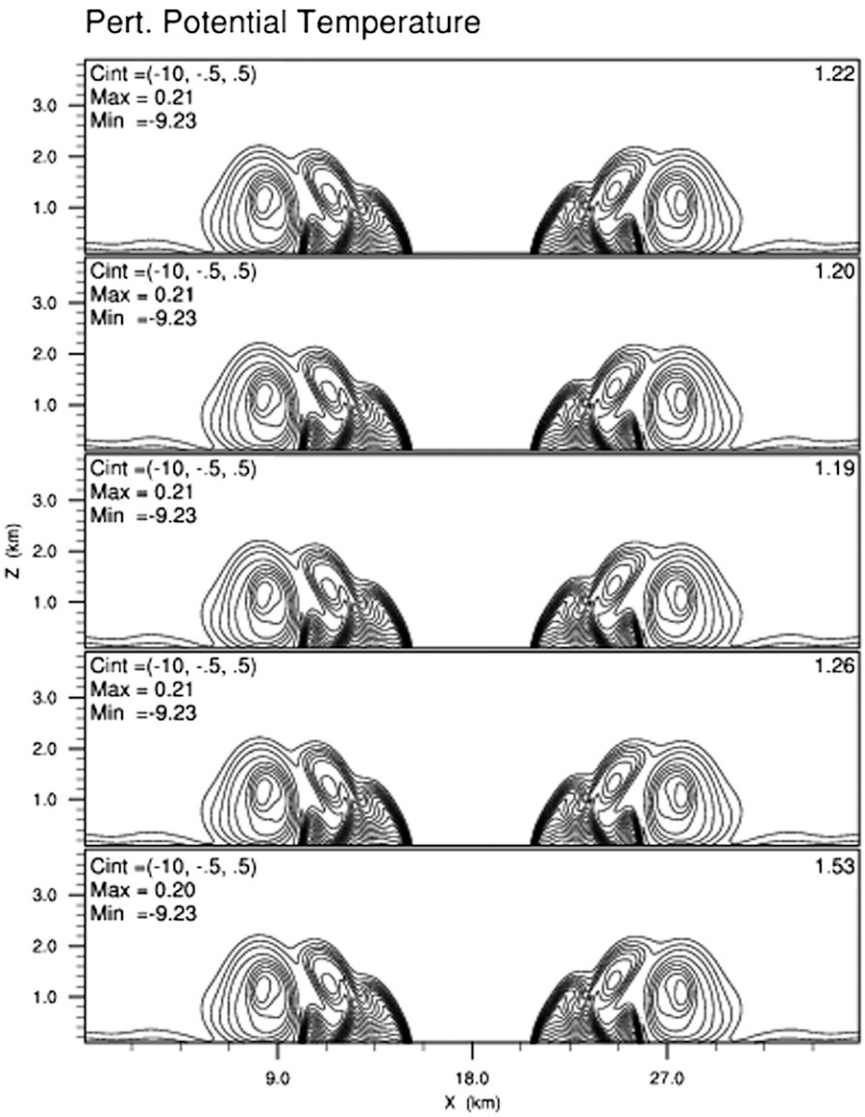

FIG. 11. Contour plot of perturbation potential temperature at $t = 900$ s obtained with step size $\Delta t = 1$ s for (top to bottom) RK3 (with divergence damping), MIS4a, TVDMISB, MIS3C, and MIS2. Displayed levels range from -10 to 0.5 (°C) in increments of 0.5. Minimum and maximum values for perturbation potential temperature are displayed in the top left of each panel and the top right shows the deviation from symmetry.

5. Discussion and outlook

We have constructed multirate infinitesimal step splitting methods for wave-type equations typically occurring in numerical weather prediction. By generalizing the split-explicit methods, the newly constructed methods provide a substantially enlarged stability region. They have desirable scale-selective damping properties.

Further, the additional order conditions for splitting methods are satisfied. The complex structure of these order conditions suggests that a resolution of these order conditions by analytic methods is impractical, if not impossible. Thus, numerical optimization procedures have to be applied to derive methods with desirable order and stability properties. Even this is a nontrivial task. We have coupled several strategies based on genetic algorithms, underlying TVD methods, and conventional optimization procedures.

In this paper we have optimized these methods for a setting where $C_A/C_S = 1/6$. The procedure proposed here can be applied to other settings, too. An example is given by the method MIS4a, which has been optimized for the case $C_A/C_S = 1/4$.

The connection between the shape of the stability region and the choice of coefficients is currently not well

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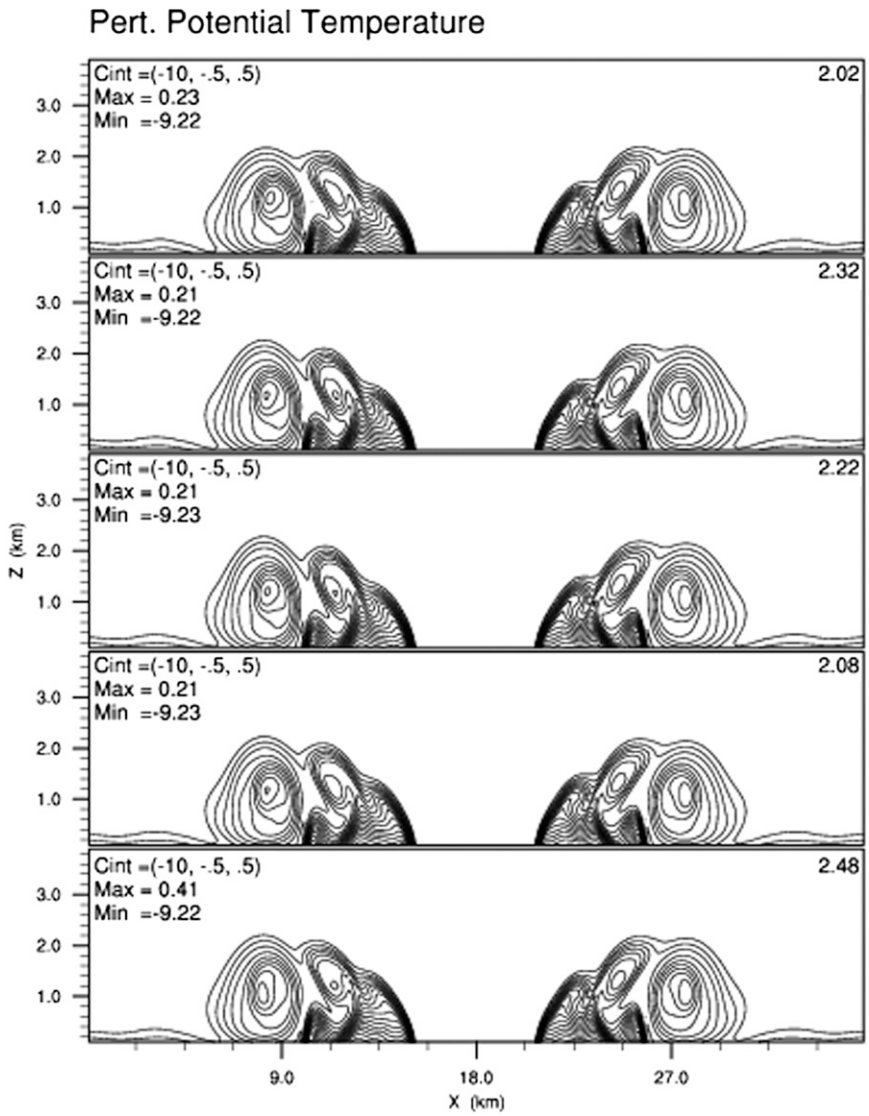

FIG. 12. As in Fig. 11, but for a step size $\Delta t = 3$ s.

understood. It is even surprising that an optimization with six free parameters (this is the case for the TVD-based methods) produces methods with a stability region quite close to the prescribed region. Further, it is not clear how much the stability region changes if the fast process integrator is replaced by an implicit method or if the spatial discretization of the fast or the slow waves is changed. The stability properties have to be checked for each choice of discretization techniques separately. A general theory for these questions seems out of reach at the moment.

Future research focuses on methods where the more restrictive ratio $C_A/C_S = 1/4$ is used or where implicit fast process integrators are applied. Finally, an open problem is the question whether methods with appropriate order and stability properties and $D = \sum_i d_i = 1$ may exist.

APPENDIX

Optimality of the Underlying TVD-RK Method

We want the underlying Runge–Kutta method of our splitting scheme to have the TVD property. To speed up the optimization process we want to fix the TVD scheme in advance. Besides good stability properties of the method we want to minimize the total fast process integration interval $D = \sum_i d_i$.

The method with optimal TVD radius is the method of Gottlieb and Shu (see Table A1). Unfortunately, we have $d_2 = c_2 = 1$ in this case, thus $D = 1 + d_3 + d_4$. In order obtain D values close to one we use a smaller value of c_2. The parameter choice $c_2 = 2/3$ seems to be a good compromise. This implies that $c_3 = 2/3$ and $b_1 = 1/4$. There

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TABLE A1. (left) TVD method of Gottlieb and Shu and (right) the method with $c_2 = 2/3$.

0				0			
1	1			2/3	2/3		
1/2	1/4	1/4		2/3	a_{31}	a_{32}	
	1/6	1/6	2/3		1/4	b_2	b_3.

is a further advantage of the choice $c_2 = 2/3$: in general the coefficients of a method with three stages are determined when c_2, c_3 are fixed. In the case $c_2 = c_2 = 2/3$ there is a free parameter left, namely, b_3.

We seek the method with maximal TVD step size among all methods with underlying RK matrix $\mathbf{A}$ given by the Butcher tableau shown in Table A1 where the parameter $c_2 = 2/3$ is fixed.

We use the standard representation of TVD methods in this section (i.e., we use coefficients $\boldsymbol{\alpha} = \alpha_{ij} \geq 0$, $\boldsymbol{\beta} = \beta_{ij} \geq 0$, where α_{ij} and $\beta_{ij} = 0$ for $i < j$). Although these coefficients correspond to the coefficients in our splitting scheme, they are not identical with the coefficients in our splitting scheme. These coefficients are not uniquely determined even if the method is defined by the RK matrix $\mathbf{A}$. An RK method in TVD form is given by

$$Y_1 = y_n, \tag{A1}$$

$$Y_i = \sum_j \alpha_{ij} Y_j + h \sum_j \beta_{ij} f(Y_j), \tag{A2}$$

where $\sum_j \alpha_{ij} = 1$ for $i \geq 2$. The TVD coefficients $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are easily related to the RK coefficients $\mathbf{A} = (a_{ij})$:

$$\mathbf{Y} = \mathbf{1}y_n + \boldsymbol{\alpha}(\mathbf{Y} - \mathbf{1}y_n) + h\boldsymbol{\beta}F(\mathbf{Y}) \Rightarrow, \tag{A3}$$

$$\mathbf{Y} = \mathbf{1}y_n + h(\mathbf{I} - \boldsymbol{\alpha})^{-1}\boldsymbol{\beta}F(\mathbf{Y}), \tag{A4}$$

where $\mathbf{Y}$, $F(\mathbf{Y})$ are the column vectors of Y_i, $F(Y_i)$. We, thus, have $\mathbf{A} = (\mathbf{I} - \boldsymbol{\alpha})^{-1}\boldsymbol{\beta}$ or $\boldsymbol{\beta} = (\mathbf{I} - \boldsymbol{\alpha})\,\mathbf{A}$, respectively.

Let the coefficients $\mathbf{A} = (a_{ij})$ (and thus the method at all) be prescribed. We choose arbitrary $\alpha_{ij} \geq 0$ to obtain β_{ij}. The method has an admissible TVD radius r if α_{ij}, $\beta_{ij} \geq 0$, where r is determined by

$$r = \min_{ij:\beta_{ij}>0} \frac{\alpha_{ij}}{\beta_{ij}}. \tag{A5}$$

We emphasize that r depends on the actual choice of admissible α, β values. Thus, the TVD radius is determined as the maximum of all admissible TVD radii.

The TVD property with TVD radius r implies: if an explicit Euler step is TVD with step size $h \leq H$, the TVD method is TVD for $h < rH$. Our method has TVD radius $r = 3/4$, which is only slightly below the radius $r = 1$ of the optimal TVD method from Gottlieb and Shu (Gottlieb and Shu 1998).

We observe first that $\mathbf{A} = (\mathbf{I} - \boldsymbol{\alpha})^{-1}\boldsymbol{\beta}$ implies $a_{ij} \geq 0$.

To determine optimal coefficients we substitute $\boldsymbol{\beta} = \mathbf{A} - \boldsymbol{\alpha}\mathbf{A}$ into the 12 inequalities $0 \leq \beta_{ij} \leq \alpha_{ij}/r$, $i = 2, 3, 4$, $j = 1, \ldots, i - 1$. We then determine coefficients α_{ij} such that the resulting upper bound on r is maximized under the additional constraints $\sum_j \alpha_{ij} = 1$, $0 \leq \alpha_{ij} \leq 1$, $\sum_j a_{ij} = c_i$, and $0 \leq a_{ij} \leq c_i$.

From stage $i = 2$ we obtain $r \leq \alpha_{21}/\beta_{21} = 3/2$, which poses no restriction on the coefficients because the maximal TVD radius is $r = 1$.

From stage $i = 3$ we obtain the following inequalities:

$$0 \leq \beta_{31} = a_{31} - 2/3\alpha_{32} \leq \alpha_{31}/r, \tag{A6}$$

$$0 \leq \beta_{32} = a_{32} \leq \alpha_{32}/r. \tag{A7}$$

The left inequality in Eq. (A6) implies $\alpha_{32} \leq 3/2a_{31}$. The choice $\alpha_{32} \leq 3/2a_{31}$ implies that the left inequality in Eq. (A6) is always satisfied. Further, it maximizes the upper bound on r resulting from Eq. (A7). Thus, the remaining bound is equivalent to

$$r \leq \frac{3}{2}\frac{a_{31}}{a_{32}} = \frac{1}{a_{32}} - 3/2. \tag{A8}$$

From stage $i = 4$ we obtain the following inequalities:

$$0 \leq \beta_{41} = 1/4 - 2/3\alpha_{42} - a_{31}\alpha_{43} \leq \alpha_{41}/r, \tag{A9}$$

$$0 \leq \beta_{42} = b_2 - a_{32}\alpha_{43} \leq \alpha_{42}/r, \tag{A10}$$

$$0 \leq \beta_{43} = b_3 \leq \alpha_{43}/r. \tag{A11}$$

The left inequality of Eq. (A10) and the right inequality of Eq. (A11) result with $b_3 a_{32} c_2 = 1/6$, $b_2 + b_3 = 3/4$ in

$$r \leq 3 - \frac{1}{a_{32}}. \tag{A12}$$

Equations (A8) and (A12) imply

$$r \leq 3/4, \tag{A13}$$

where $r = 3/4$ implies $a_{32} = 4/9$, $a_{31} = 2/9$, $b_3 9/16$, and $b_2 = 3/16$. The matrix $\boldsymbol{\alpha}$ is not uniquely determined, we have $\alpha_{32} = 1/3$, $\alpha_{43} = 27/64$, and $\alpha_{42} \leq 15/64$. With $\alpha_{42} = 0$ we obtain the TVD scheme:

$$\begin{aligned} Y_2 &= y_n + 2/3hf(y_n) \\ Y_3 &= 2/3y_n + 1/3Y_2 + 4/9f(Y_2) \\ y_{n+1} &= 37/64y_n + 27/64Y_3 + 5/32hf(y_n) + 9/16hf(Y_3). \end{aligned} \tag{A14}$$

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The corresponding Butcher tableau is given in Table 5. It has TVD radius of $r = 3/4$, which is slightly smaller than the optimum value. Nevertheless, because the node $c_2 = 2/3$ is smaller than the node $c_2 = 1$ of the method of Gottlieb and Shu it is more suitable for the construction of MIS-RK methods.

REFERENCES

Baldauf, M., 2010: Linear stability analysis of Runge–Kutta-based partial time-splitting schemes for the Euler equations. *Mon. Wea. Rev.,* **138,** 4475–4496, doi:10.1175/2010MWR3355.1.

Durran, D. R., and P. N. Blossey, 2012: Implicit-explicit multistep methods for fast-wave–slow-wave problems. *Mon. Wea. Rev.,* **140,** 1307–1325, doi:10.1175/MWR-D-11-00088.1.

Gottlieb, S., and C.-W. Shu, 1998: Total variation diminishing Runge–Kutta schemes. *Math. Comput.,* **67,** 73–85, doi:10.1090/S0025-5718-98-00913-2.

Jebens, S., O. Knoth, and R. Weiner, 2009: Explicit two-step peer methods for the compressible Euler equations. *Mon. Wea. Rev.,* **137,** 2380–2392, doi:10.1175/2008MWR2671.1.

Knoth, O., and R. Wolke, 1998: Implicit–explicit Runge–Kutta methods for computing atmospheric reactive flows. *Appl. Numer. Math.,* **28,** 327–341, doi:10.1016/S0168-9274(98)00051-8.

Purser, R. J., 2007: Accuracy considerations of time-splitting methods for models using two-time-level schemes. *Mon. Wea. Rev.,* **135,** 1158–1164, doi:10.1175/MWR3339.1.

Skamarock, W., and J. Klemp, 1992: The stability of time-split numerical methods for the hydrostatic and the nonhydrostatic elastic equations. *Mon. Wea. Rev.,* **120,** 2109–2127, doi:10.1175/1520-0493(1992)120<2109:TSOTSN>2.0.CO;2.

Straka, J., R. Wilhelmson, L. Wicker, J. Anderson, and K. Droegemeier, 1993: Numerical solutions of a non-linear density current: A benchmark solution and comparisons. *Int. J. Numer. Methods Fluids,* **17,** 1–22, doi:10.1002/fld.1650170103.

Vater, S., R. Klein, and O. Knio, 2011: A scale-selective multilevel method for long-wave linear acoustics. *Acta Geophys.,* **59,** 1076–1108, doi:10.2478/s11600-011-0037-x.

Wensch, J., O. Knoth, and A. Galant, 2009: Multirate infinitesimal step methods for atmospheric flow simulation. *BIT Numer. Math.,* **49,** 449–473, doi:10.1007/s10543-009-0222-3.

Wicker, L. J., 2009: A two-step Adams–Bashforth–Moulton split-explicit integrator for compressible atmospheric models. *Mon. Wea. Rev.,* **137,** 3588–3595, doi:10.1175/2009MWR2838.1.

——, and W. Skamarock, 1998: A time-splitting scheme for the elastic equations incorporating second-order Runge–Kutta time differencing. *Mon. Wea. Rev.,* **126,** 1992–1999, doi:10.1175/1520-0493(1998)126<1992:ATSSFT>2.0.CO;2.

——, and ——, 2002: Time splitting methods for elastic models using forward time schemes. *Mon. Wea. Rev.,* **130,** 2088–2097, doi:10.1175/1520-0493(2002)130<2088:TSMFEM>2.0.CO;2.

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